

SOFTWARE REQUIREMENT SPECIFICATION (SRS) DOCUMENT

PRESENTED BY:	PRESENTED TO:	DATE:
ANF.co	TS. Azma binti Abdullah	12/01/2024

TEAM MEMBERS

Chief Executive Officer	Chief Business Analyst	Software Quality Assurance
		
NURUNNAJIHAH BINTI ZAINAL CB23027	MOHAMAD FAHMI BIN MOHD RAZALI CB23021	NURUL AIN AZIMAH BINTI AZHAR CB23105

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Faculty of Computing, Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA)

SOFTWARE REQUIREMENT SPECIFICATION (SRS) FACULTY OF COMPUTING

DOCUMENT APPROVAL

	Name	Date
Authenticated by:  _____ Project Leader	Nurunnajihah binti Zainal	
Approved by: _____ Project Manager	TS. Azma binti Abdullah	
Approved by: _____ Client	Puan Siti Rafidah binti Ahmad Fuad	

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LIST OF ABBREVIATIONS

1. INTRODUCTION

1.1 PURPOSE

The purpose of this document is to provide a detailed description for the News Check System which focuses on its requirements and how it will help the public in verifying news. We aim to understand the functionalities of the system and how the system will be used. Additionally, it describes the system's objectives, specifications, including identification and interfaces for each functional requirement. These documents must be presented to the clients for approval and to serve as a reference for the development team to fully understand the system's requirements. (Binding contract with developer/stakeholder)

1.2 SYSTEM IDENTIFICATION

System Title	News Check System	
System Abbreviation	NCS-2024	
System Identification Number	NCS_2024-1	
	Meanings for terms use:	
	NCS	News Check System (System Name)
	2024	Year document created
System Version Number	1	Version of release
	1.0.24	
	This is the initial release for the News Check System to the user.	
	1.0.24	
System Release Number	- Starting with number 1 as the major release. - Middle with number 0 as the minor release number that this is initial version of the system. - Number 24 at the service releases shows that this system is released in 2024.	
Document Identification ID	SRS-NCS-2024	
	Meanings for terms use:	
	SRS	Software Requirement Specification
	NCS	News Check System (System Name)

	<table> <tr> <td>2024</td><td>Year document created</td></tr> </table>	2024	Year document created						
2024	Year document created								
Use Case ID	UC100-NCS-2024 Meanings for terms use: <table> <tr> <td>UC</td><td>Use Case</td></tr> <tr> <td>100</td><td>Number of use case in this system</td></tr> <tr> <td>NCS</td><td>News Check System (System Name)</td></tr> <tr> <td>2024</td><td>Year document created</td></tr> </table>	UC	Use Case	100	Number of use case in this system	NCS	News Check System (System Name)	2024	Year document created
UC	Use Case								
100	Number of use case in this system								
NCS	News Check System (System Name)								
2024	Year document created								
Requirement ID	RQ101-NCS-2024 Meanings for terms use: <table> <tr> <td>RQ</td><td>Requirement</td></tr> <tr> <td>100</td><td>Number of requirements in this system</td></tr> <tr> <td>NCS</td><td>News Check System (System Name)</td></tr> <tr> <td>2024</td><td>Year document created</td></tr> </table>	RQ	Requirement	100	Number of requirements in this system	NCS	News Check System (System Name)	2024	Year document created
RQ	Requirement								
100	Number of requirements in this system								
NCS	News Check System (System Name)								
2024	Year document created								

1.3 SYSTEM OVERVIEW

The News Check system is designed to help users in verifying the legitimacy of news articles, social media posts, and other forms of information to minimize the spread of false information. The system acts as a reliable tool by checking articles, sources and data for factual accuracy and reliability. The software of the system involves various users which are public user, administrators and management approver and agency. The system is divided into three modules, each addressing a specific aspect of the verifying news process:

Module 1: Comprehensive profile data for all users.

This module stores detailed profile data for all users, including public user, management approver, administrator and agencies. For public user, they are required to register their contact details, including first name, last name, email address, phone number, username and password. They can subsequently log in to the system using their registered credentials. For administrator, they access the system by using a default username and password to monitor system activity. Management approver and agencies receive their assigned usernames and passwords by the administrator, which they use to login the system. Comprehensive profile data for all users can provide a range of valuable reports depending on the data categories obtained, including total logins over a period, successful logins and failed logins of the user along with the percentage.

Module 2: Capability to submit forms for the verification of news validity.

This module allows public users to report unverified or suspicious news for verification. Public users enter details such as subject, select category, description, news URL, source of the news and evidence before submitting the form. They need to upload relevant evidence such as screenshot comments from the news platform to support their submission. For the category selection, this is to enable user to select the appropriate category such as disaster, economy, education, crime and health. By using this category selection, administrator can quickly assign the submission to the relevant agencies depending on the report. Management approver needs to review the news submission that was submitted by the public user to ensure the details given are correct. For

Agencies, they can view list of the news submission from the public user. Administrators can view the list of the news submission and generate reports showing the list of news forms submitted. These reports will include the submission id, news category, subject, submission date and assigned agency.

Module 3: Classification of complaint forms forwarded to the appropriate authorities.

This module provides for classifying complaints or submissions into appropriate agency categories. Once a public user submits a report, the system analyzes the given details such as subject, select category, description, news URL, news source and evidence. Based on this input, the administrator is assigned to submit the complaint to the relevant agency for further action. Administrators need to see each submission news and classify it by selecting the existing agency in the system. Before the administrator classifies the submission news to the agency, the Management Approver needs to check the submission news first and give a pass or reject response after the public users send their submission news. For agency assignments, they can see all the submission news after the administrator classified the submission news form. Then the new agency can do any response to all submission news that involves their agency. For reports, administrators can generate all news submission reports that are sent the most by public users according to agency.

1.4 REFERENCES

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3. TS. AZMA BINTI ABDULLAH (2023). Appendix 2: SRS Template Document Student. Pekan, Pahang: Universiti Malaysia Pahang Al-Sultan Abdullah
4. SALUR KEPADA KAMI. (2017, February 28). SEBENARNYA.MY.
<https://sebenarnya.my/salur/>
5. Bailin, S. C. (1997). Software Requirements Engineering. Wiley-IEEE Computer Society Press.

2. PRODUCT DESCRIPTION

2.1 PRODUCT PERSPECTIVE

Figure 2.1 shows the context diagram of the News Check System. This figure illustrates the overall system functions and interactions within the system. The system involves four external entities which are public person, administrator, management approver and agency.

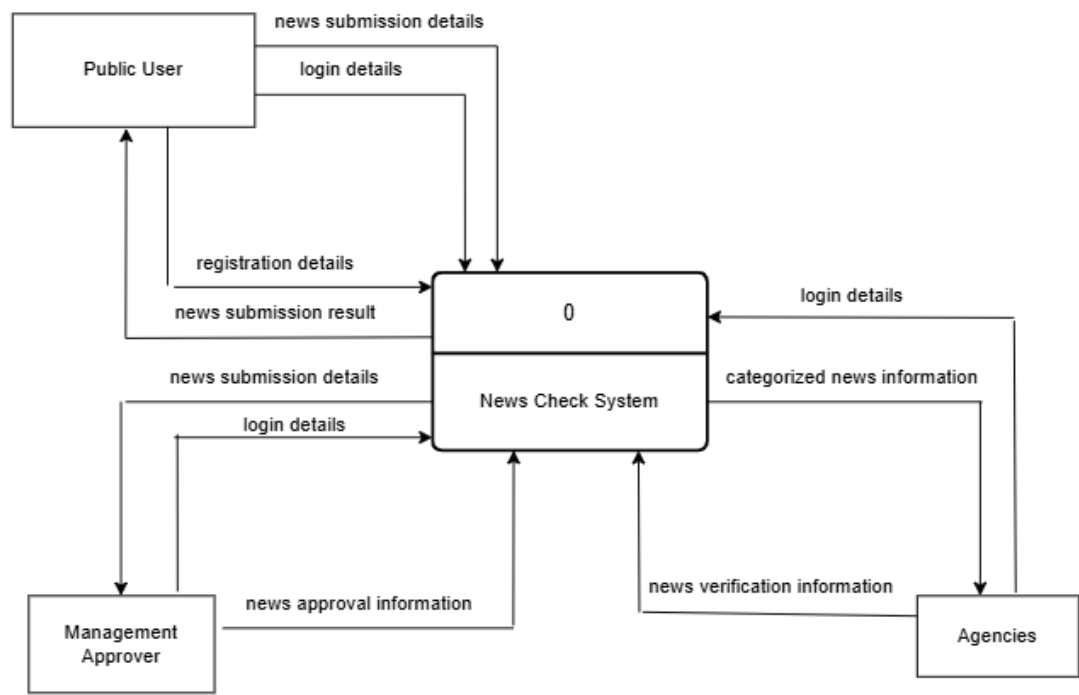


Figure 2.1 Context Diagram of News Check System

2.2 PRODUCT FUNCTION

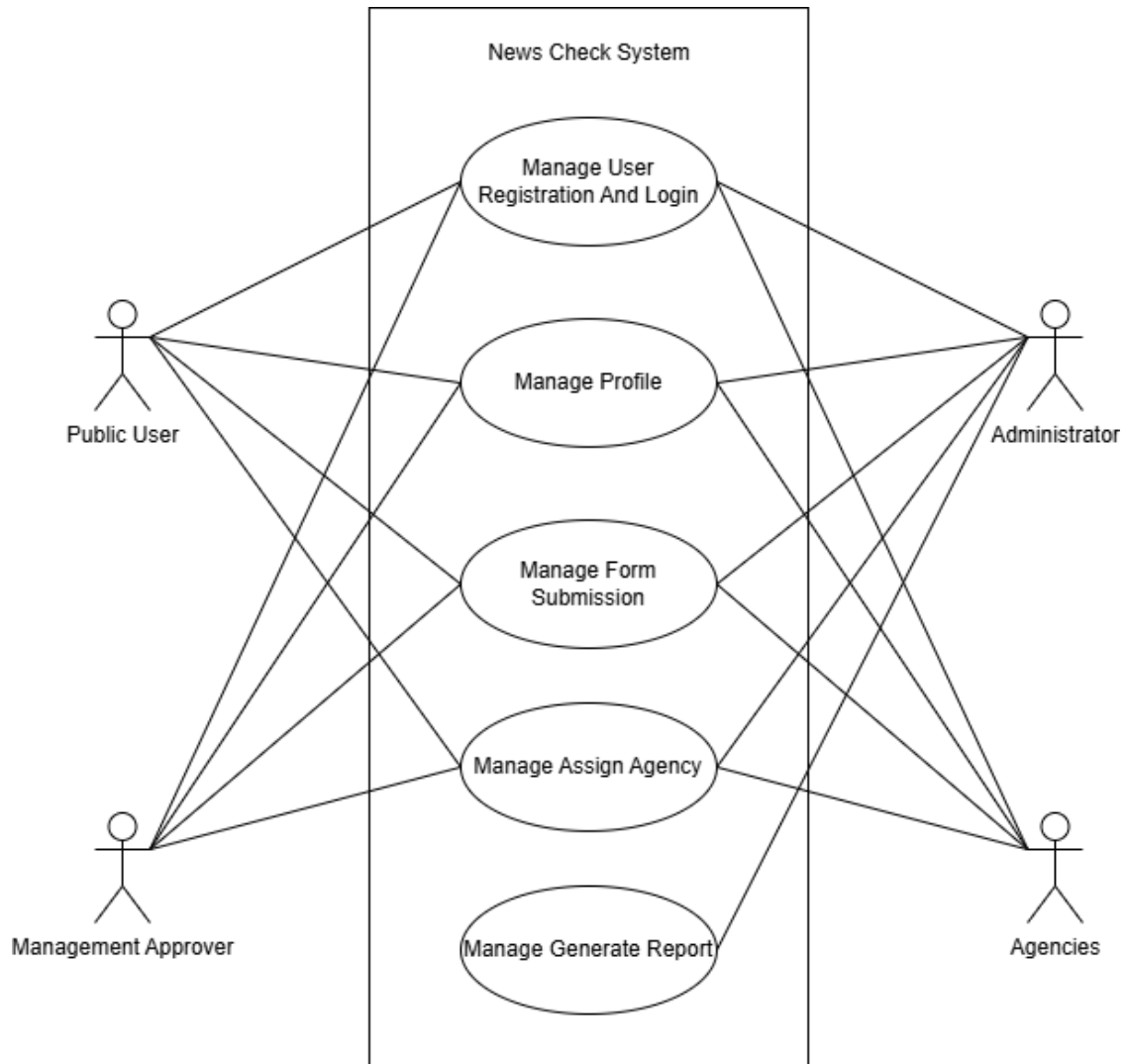


Figure 2.2 Use Case Diagram of News Check System

2.3 USER CHARACTERISTICS

The News Check System (NCS) has four main users:

User	Educational Level	Experience	Technical Expertise
Public User	Basic education to higher education.	Have experience with similar systems.	Proficient in using basic internet and capable of filling in verifying news forms.
Administrator	At least diploma or bachelor's degree in IT, computer science or related field.	Know how to manage system operation and digital platform.	Proficient in database management and monitoring system workflow.
Management Approver	At least diploma or bachelor's degree in public administrator or related field.	Experience in decision-making, reviewing and approving processes.	Proficient in navigating the system to review and approve reports.
Agencies (e.g. PDRM, KKM, KPM, JBPM, MOF)	Bachelor's degree or training in their specialized field relevant to the agencies	Experience in analyzing the verifying news form and can verify the news based on data from their agencies.	Proficient in investigating digital evidence to verify the authenticity of news sources.

3. SPECIFIC REQUIREMENTS (SOFTWARE PRODUCT FEATURES)

3.1 MANAGE ASSIGN AGENCY USE CASE

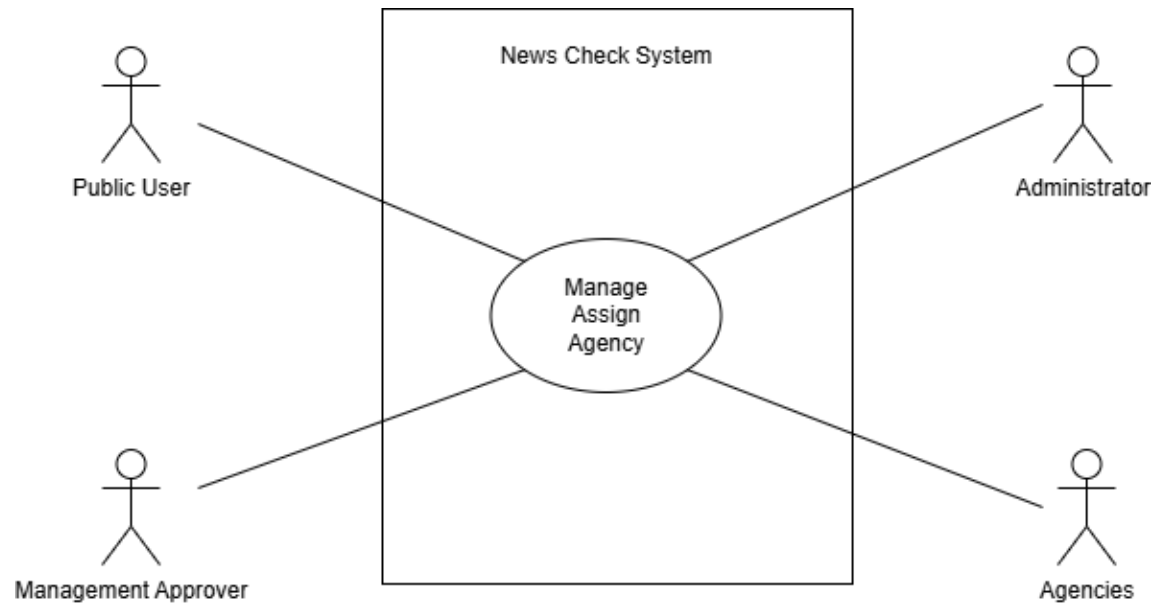


Figure 3.1 Manage Assign Agency-Use Case

3.1.1 Use Case Description for Manage Assign Agency

Table 3.1 Manage Assign Agency -Use Case Description

Use Case ID	UC400-NCS-2024
Brief Description	This use case allows the Administrator to assign agencies to submissions provided by the Public User, enabling them to handle specific tasks or news verification. The system ensures that all assignments are reviewed and approved by the Management Approver, providing the assigned agencies with task details and tracking their progress until completion.
Actor	Public user, Administrator, Management Approver and Agencies
Pre-Conditions	1. The all users must be logged into the system as their respective roles.
Basic Flow	The use case begins with: Public User 1. The Public User to the <<News Submission>> section and selecting << Submission History >>. [RQ401-NCS] 2. The system displays “Submission History” including submission id, new category, submission date, status and detail news submission. 3. The public user views the submission’s status of the form to

	which agency. - The use case ends for the public user. Administrator - The administrator navigates to the <<News Submission>> section and selecting <<List>>. [RQ402-NCS] - The system displays “List of News Submission” including submission id, news category, submission date, status, assigned agency and action buttons. - The administrator selects a news submission by clicking the <<View>> button. [RQ403-NCS] - The system displays the detailed view of the selected news submission, including fields such as subject, category, description, source, URL and evidence. - The administrator identifies that the news submission requires additional information or clarification. [A1: Request Update] - The administrator assigns the news submission to an appropriate agency by searching an agency from the dropdown menu. [E1: No Matching Agencies Found] - The administrator clicks the <<Submit>> button to forward the news to the selected agency. [RQ404-NCS] - The system notifies the agency. - The use case ends for the administrator. Management Approver - The management approver navigates to the <<News Submission>> section and <<List>>. [RQ405-NCS] - The system displays “List of News Submission” including submission id, new category, submission date, status, assigned agency and action buttons. - The management approver views the submission’s status of the form to which agency. - The use case ends for the management approver. Agencies - The agencies navigate to the <<News Submission>> section and <<List>>. [RQ406-NCS] - The system displays “List of News Submission” including submission id, subject, submission date, status, review assigned agency and action buttons. - The agencies view the submission’s status of the form news submission from public user. - The use case ends for the Agencies.
Alternative Flow	[A1: Administrator Requests Update on Submission] - The administrator determines that it requires further clarification or additional information such as missing evidence and unclear description. - The administrator selects the <<Request Update>> button.

	[RQ407-NCS] 3. The system prompts the administrator to provide a reason or note for the update request. 4. The system updates the public user submission status to "Update Requested" for the request. 5. The process continues to step 3 from basic flow of administrator flow.
Exception Flow	[E1: No Matching Agencies Found] 1. The administrator attempts to assign the submission but finds no appropriate agency in the dropdown menu. 2. The system displays an “No record Found” indicating the failure. 3. The system provides clear feedback explaining why no matching agencies are available. 4. The process continues to step 3 from basic flow of administrator flow.
Post-Conditions	1. The public user can see the status of their news submission including whether it has been assigned to an agency or if further updates are requested. 2. The administrator successfully assigns the news submission to an appropriate agency, updates its status, or requests clarification from the public user if needed. 3. The management approver is updated about the status of news submissions and can view which agency has been assigned to handle each case. 4. The assigned agency has received a notification regarding the new submission and can view the details on their "List of News Submission" page.
Rules	1. The public user can only view the status of submission news forms sent to which agency. 2. The administrator can only assign news submission to the agency.
Constraints	1. The Administrators can only request updates from public users once per submission cycle to prevent spamming.

3.1.2 Manage Assign Agency Sequence Diagram

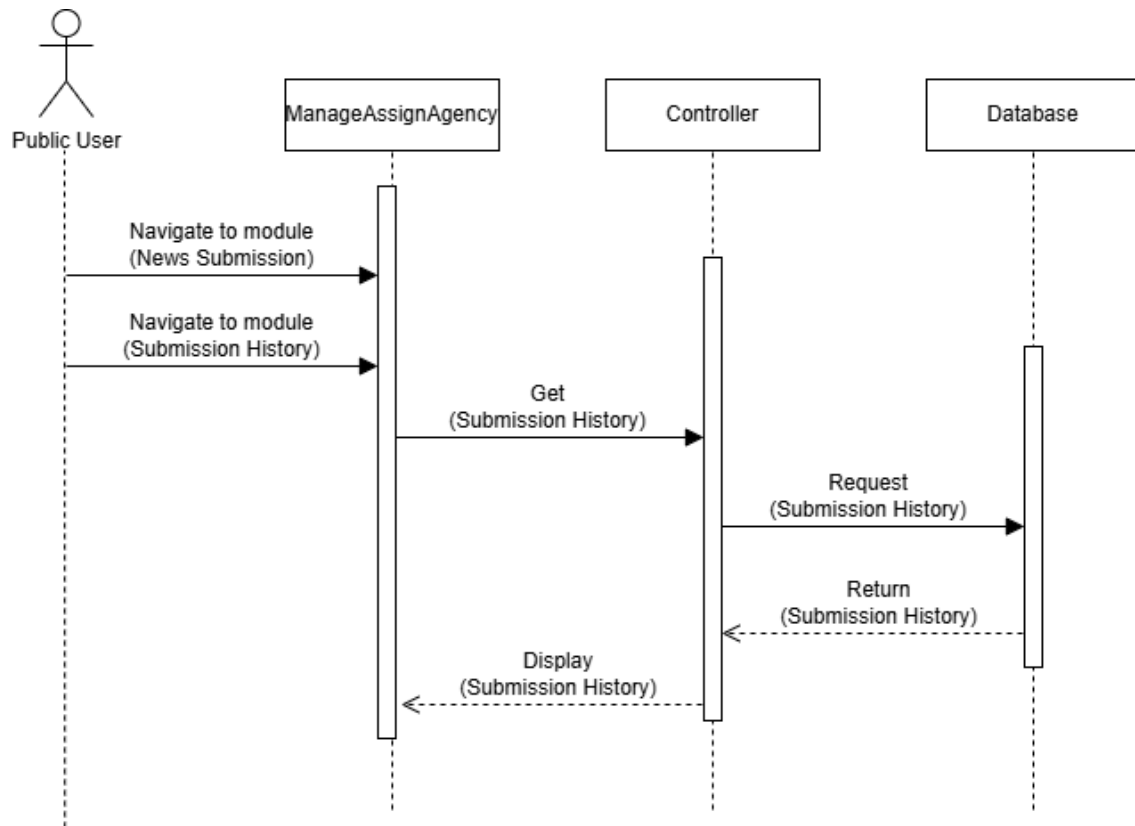


Figure 3.2 Basic Flow Public User for Manage Assign Agency

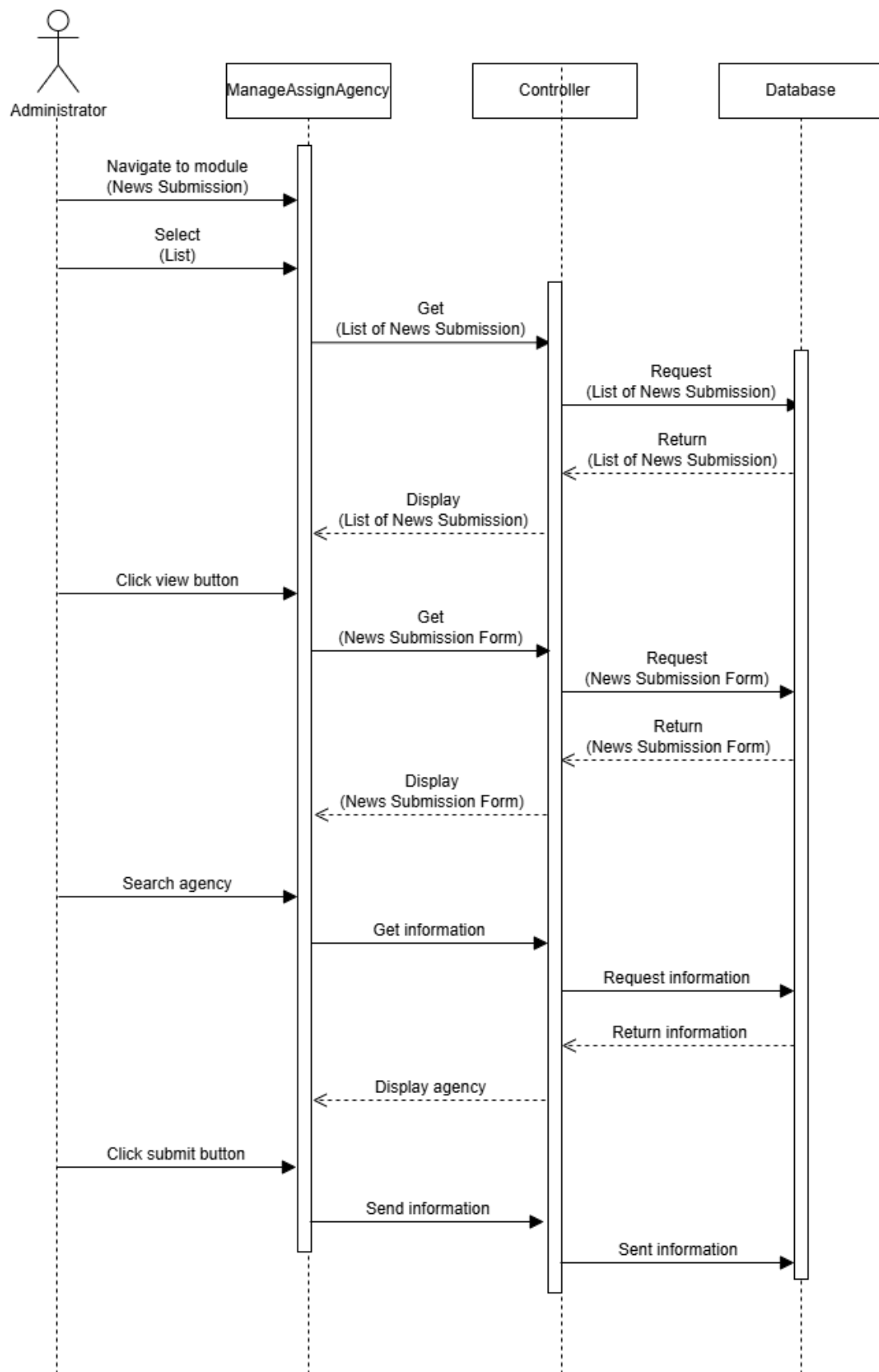


Figure 3.3 Basic Flow Administrator for Manage Assign Agency

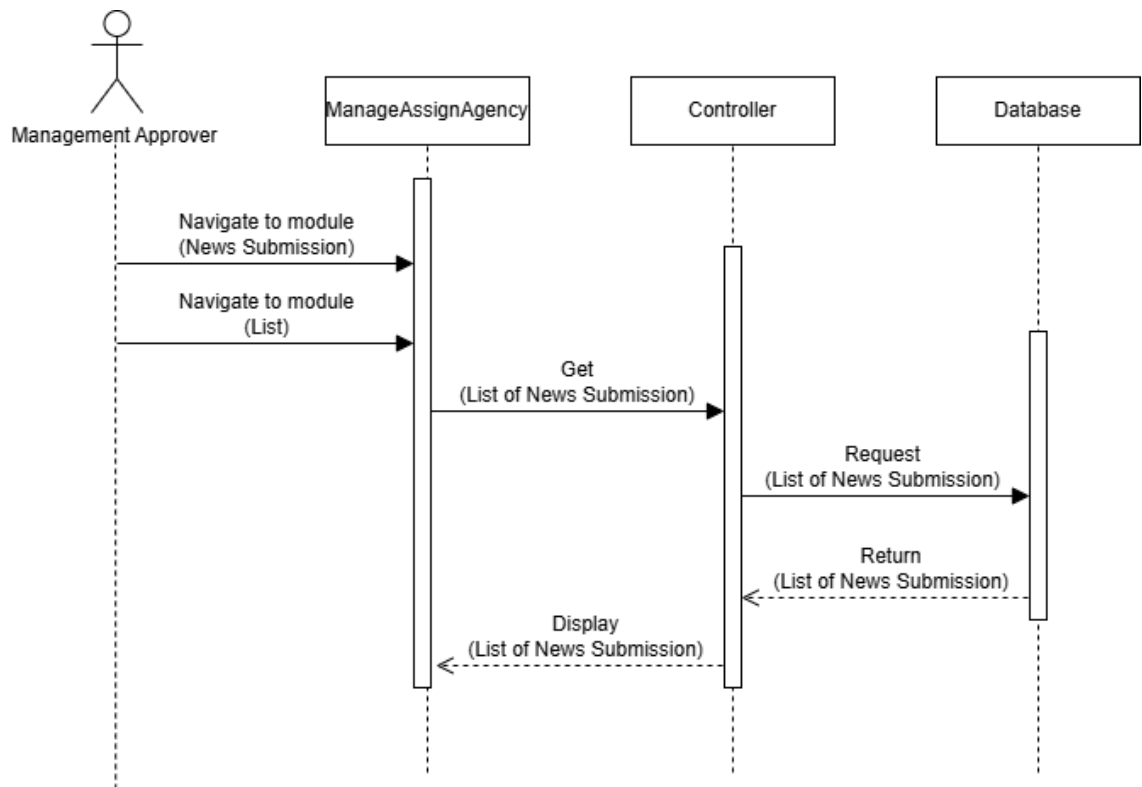


Figure 3.4 Basic Flow Management Approver for Manage Assign Agency

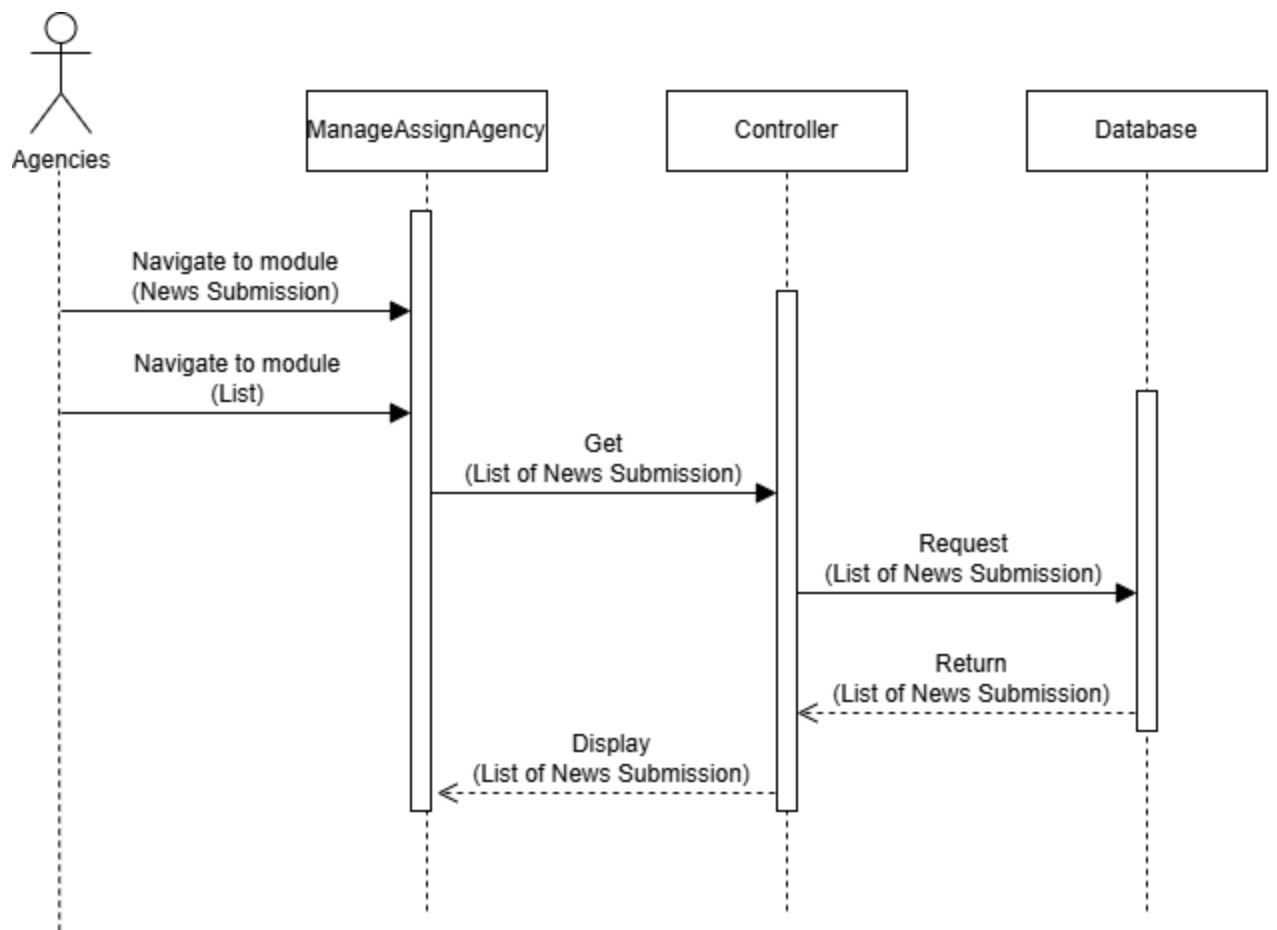


Figure 3.5 Basic Flow Agencies for Manage Assign Agency

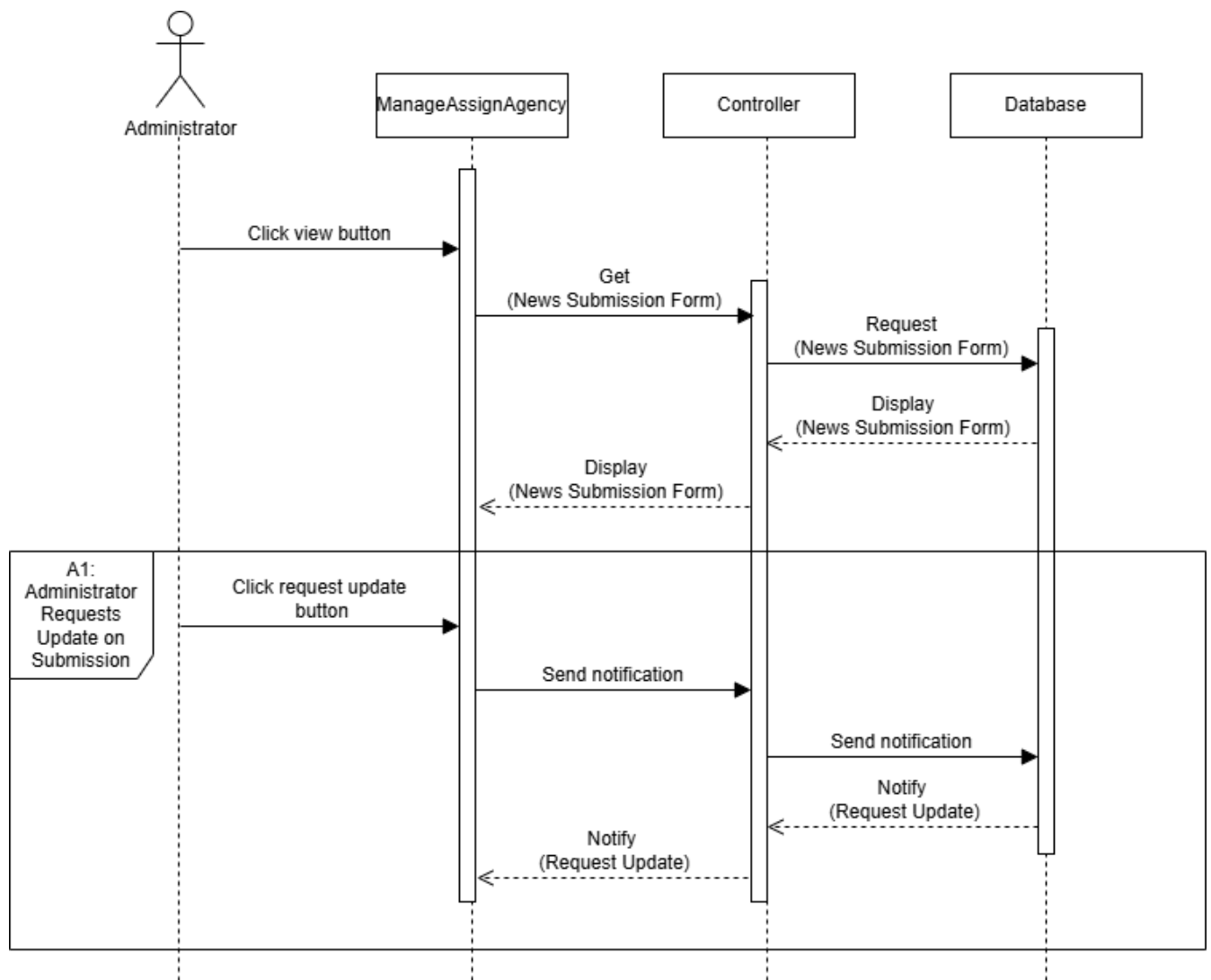


Figure 3.6 Alternative Flow Administrator for Manage Assign Agency

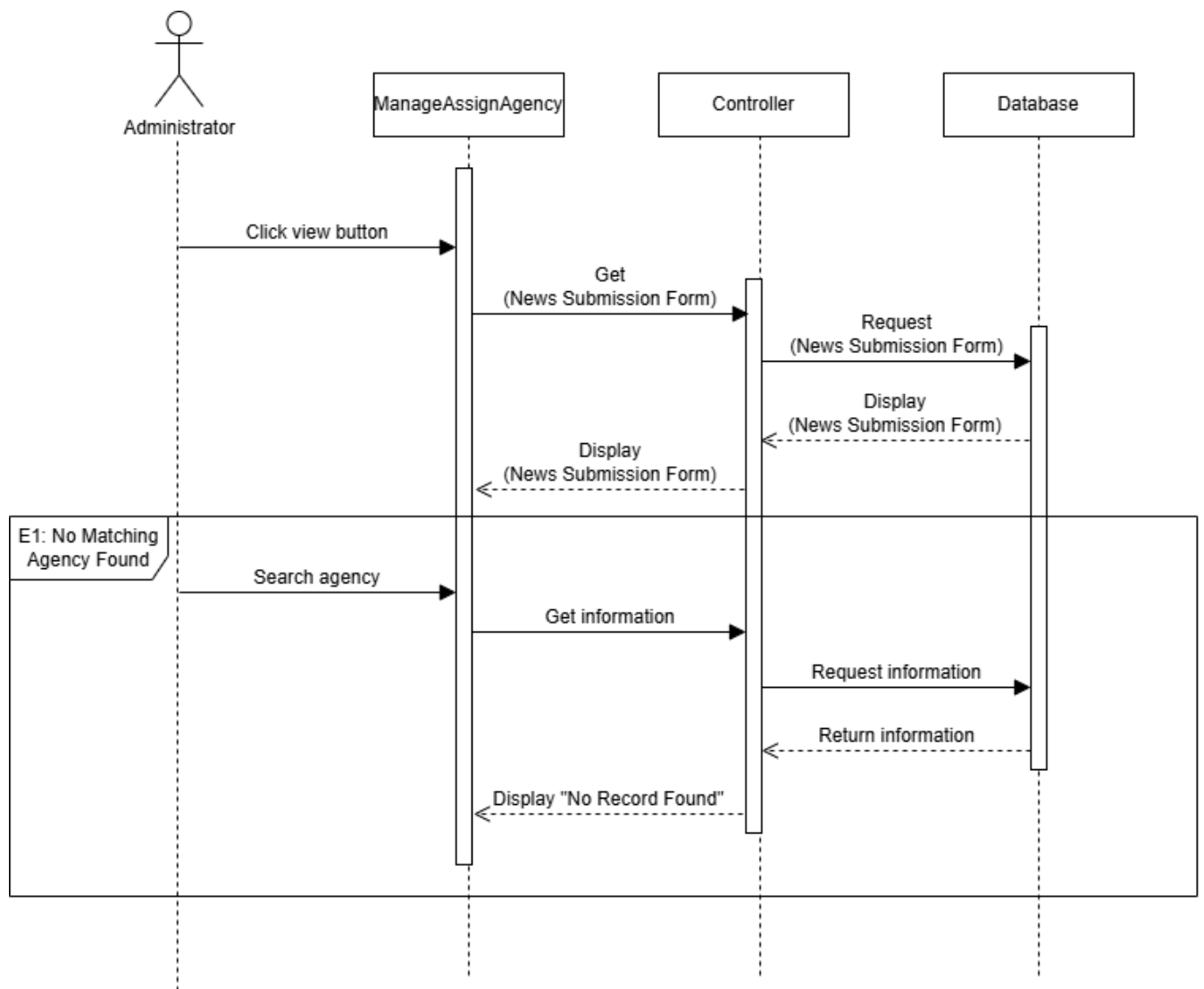


Figure 3.7 Exception Flow Administrator for Manage Assign Agency

3.1.3 Manage Assign Agency User Interface

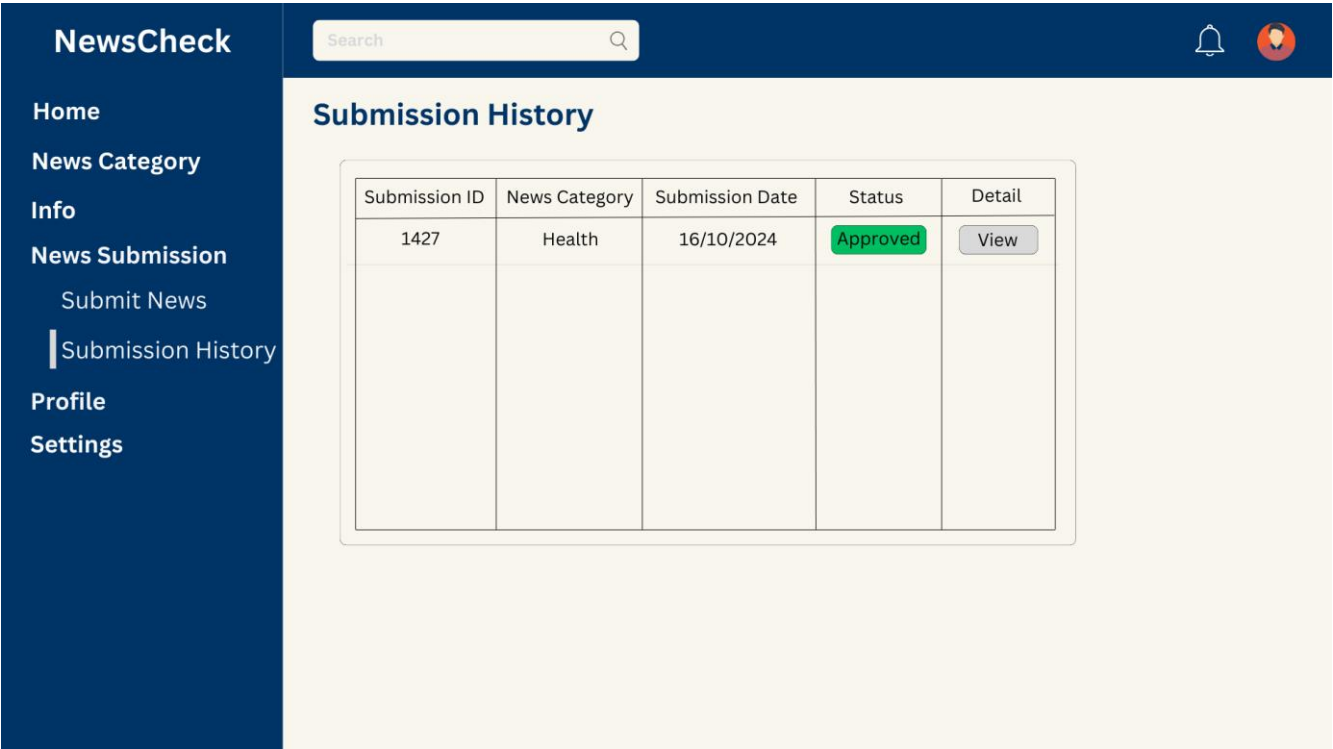


Figure 3.8 User Interface of Submission History for Public User

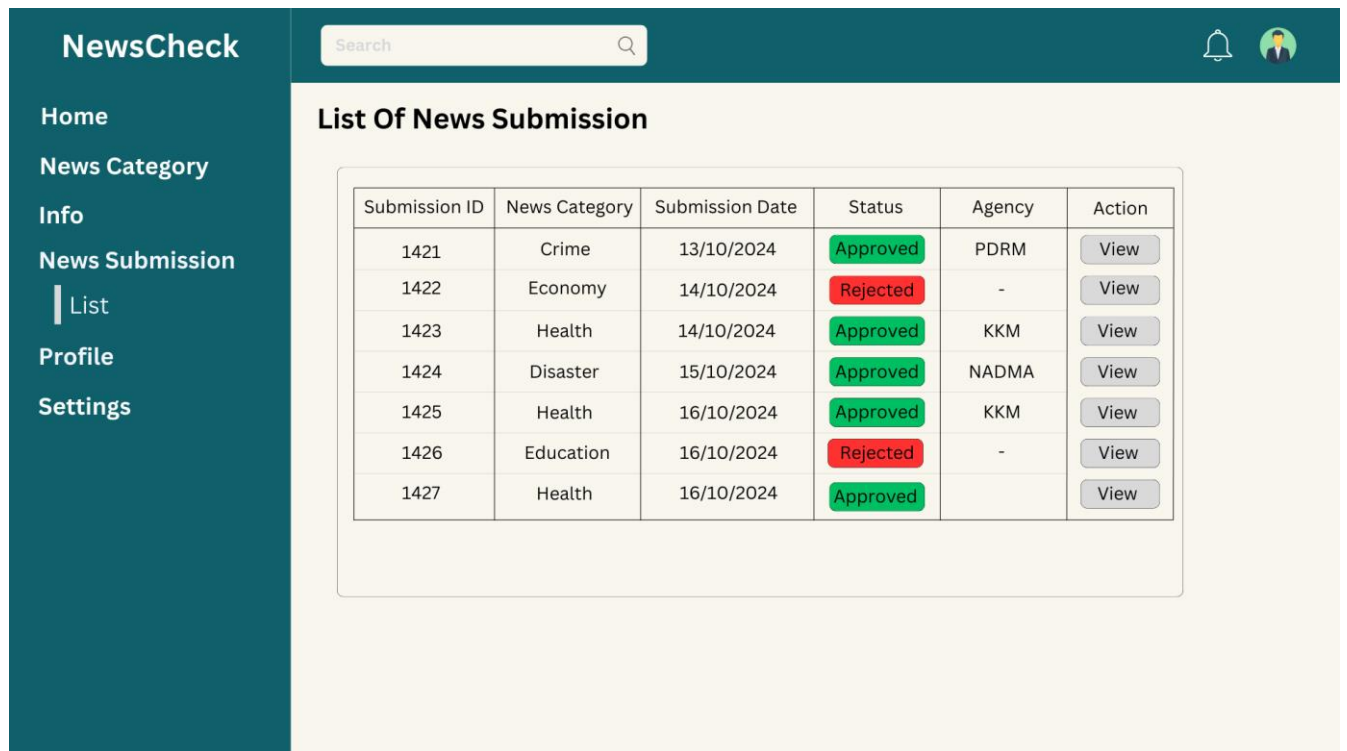


Figure 3.9 User Interface of News Submission List for Management Approver

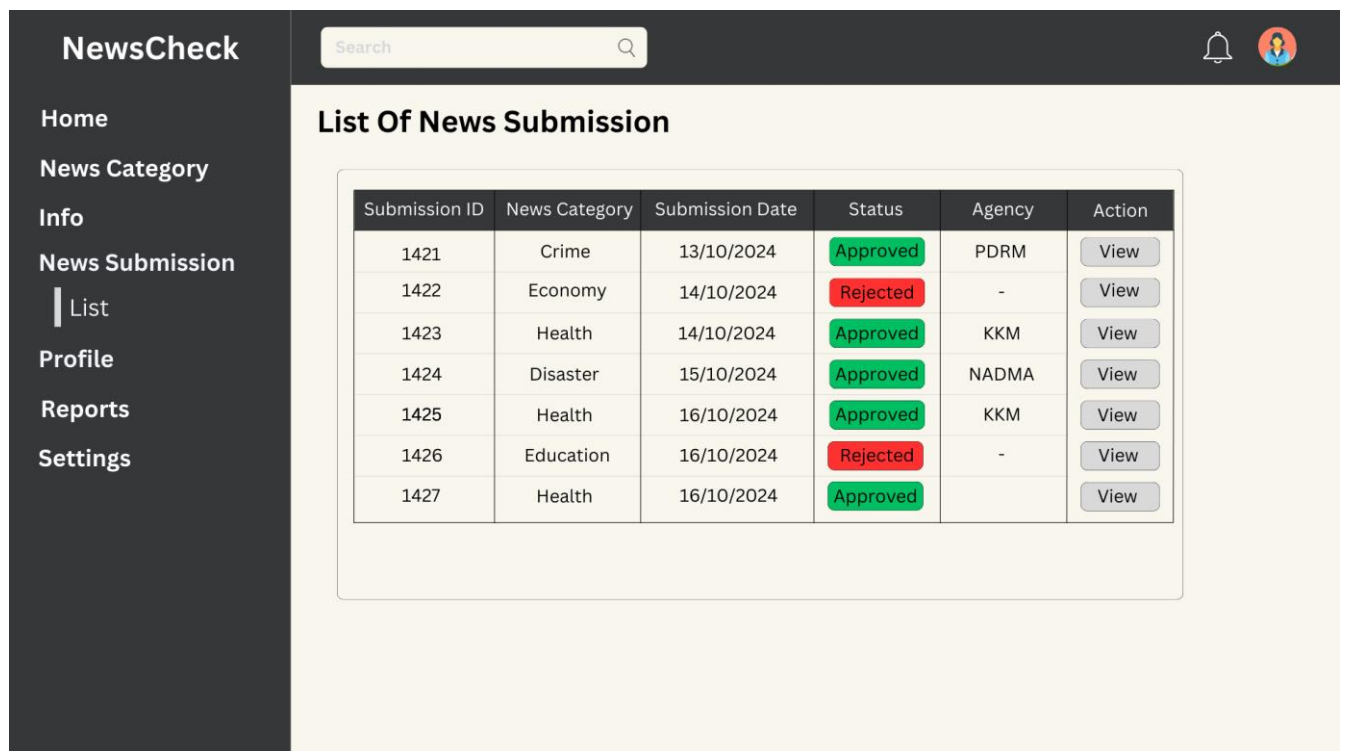


Figure 3.10 User Interface of News Submission List Before Assign Agency for Administrator

NewsCheck

Home

News Category

Info

News Submission

List

Profile

Reports

Settings

Search

Submission ID :

1427

Subject :

HMPV virus

News Category :

Health

Description :

HMPV virus already in Malaysia since last year

News URL :

https://www.facebook.com/share/p/19iFi55CsZ/

News Source :

Facebook

Date:

16/10/2024

Agency :

Search...

KKM

JPN

MDEC

Evidence: *Click To Open

Request Update

Submit

Figure 3.11 User Interface of News Submission List Form for Administrator

NewsCheck

Home

News Category

Info

News Submission

List

Profile

Reports

Settings

Search

Submission ID

News Category

Submission Date

Status

Agency

Action

1421

Crime

13/10/2024

Approved

PDRM

View

1422

Economy

14/10/2024

Rejected

-

View

1423

Health

14/10/2024

Approved

KKM

View

1424

Disaster

15/10/2024

Approved

NADMA

View

1425

Health

16/10/2024

Approved

KKM

View

1426

Education

16/10/2024

Rejected

-

View

1427

Health

16/10/2024

Approved

KKM

View

Figure 3.12 User Interface of News Submission List After Assign Agency for Administrator

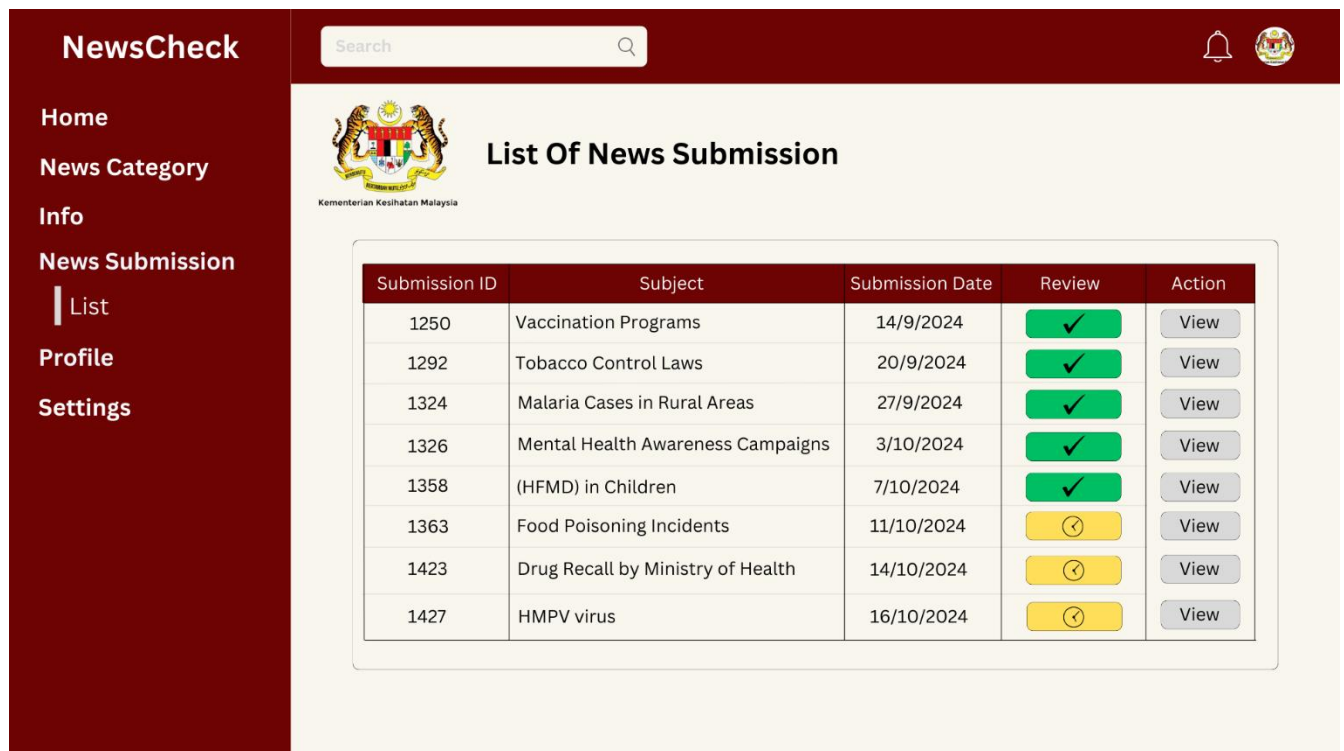


Figure 3.13 User Interface of News Submission List for Agency

Table 3.2 Manage Assign Agency - User Interface Description

User Interface Name	Description
Submission History for Public User	The Public User can view the list of news submission history in this interface.
News Submission List for Management Approver	The Management Approver can view the list of news submission in this interface.
News Submission List Before Assign Agency for Administrator	The Administrator can view the list of news submission in this interface.
News Submission List Form for Administrator	The Administrator can assign agency the news submission form in this interface by searching the agency related with the news and click submit button, it will notify the agency.
News Submission List After Assign Agency for Administrator	The Administrator can view the list of news submission in this interface that they have assign agency.
News Submission List for Agency	The Agency can view the list of news submission in this interface.

3.2 MANAGE GENERATE REPORT

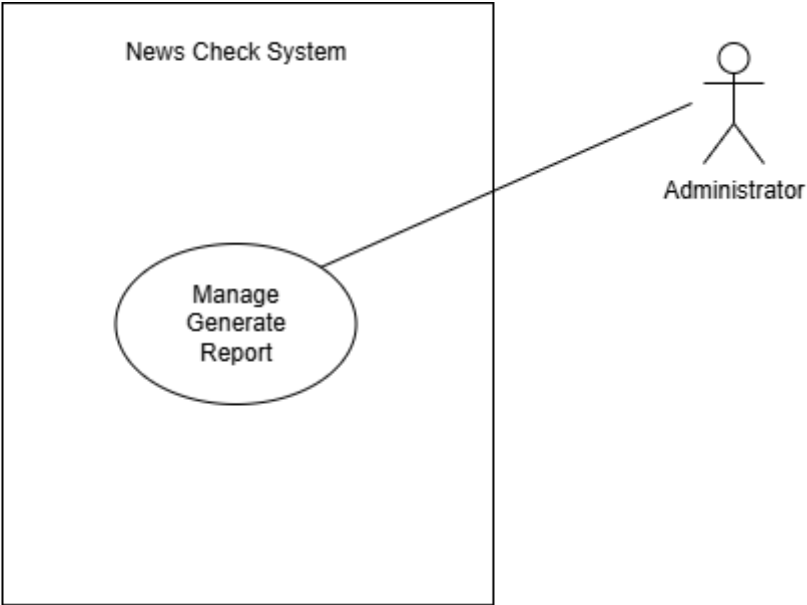


Figure 3.14 Manage Generate Report-Use Case

3.2.1 Manage Generate Report Use Case Description

Table 3.3 Manage Generate Report -Use Case Description

Use Case ID	UC500-NCS-2024
Brief Description	This use case allows the administrator to generate and view various reports in the News Check System. The functionality ensures that the administrator can generate accurate and meaningful reports to support decision-making and analysis.
Actor	Administrator
Pre-Conditions	1. The administrator must be logged into the system.
Basic Flow	1. The administrator navigates to the <<Reports>> section in the system. 2. The administrator selects a specific report type such as <<Ranking News Categories Report>>. [RQ501-NCS] 3. The system displays “Ranking News Categories Report” including news category, number of submissions, percentage of total, and rank. 4. The system prompts the administrator to specify any additional filters such as date and clicks the<<Apply>> button to filter the submissions. [A1: No Data Available for Selected Date Range] [E1: Invalid Filter Date] [RQ502-NCS] 5. The system displays information including Submission id, subject, submission date, and agency. 6. The administrator clicks <<Generate Report>> button. [RQ503-NCS]

	- The administrator may save or download it. - The use case end.
Alternative Flow	[A1: No Data Available for Selected Date Range] - The system finds no news submissions matching the selected date range. - The system displays the message: "No submissions available for the selected period." - The process continues at basic flow of number 4.
Exception Flow	[E1: Invalid Filter Date] - The administrator provides invalid input date. - The system displays an error message and prompts for corrections. - The process continues at basic flow of number 4.
Post-Conditions	A report is successfully generated and displayed.
Rules	The administrator only can access "Manage Generate Report" feature.
Constraints	The system must handle large data sets efficiently but may impose constraints on generating reports that require processing beyond system limits such as millions of records.

3.2.2 Manage Generate Report Sequence Diagram

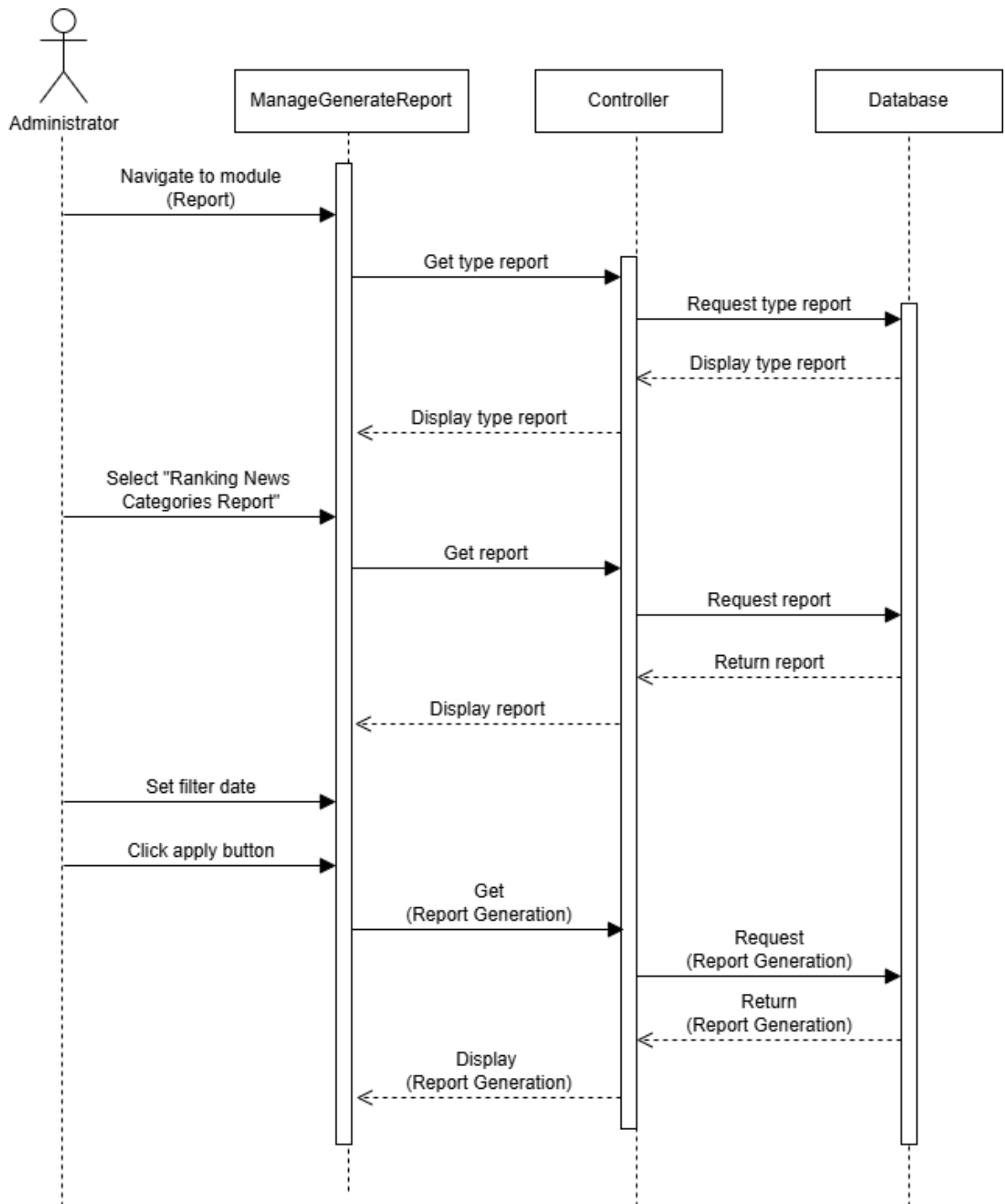


Figure 3.15 Basic Flow Administrator for Manage Generate Report

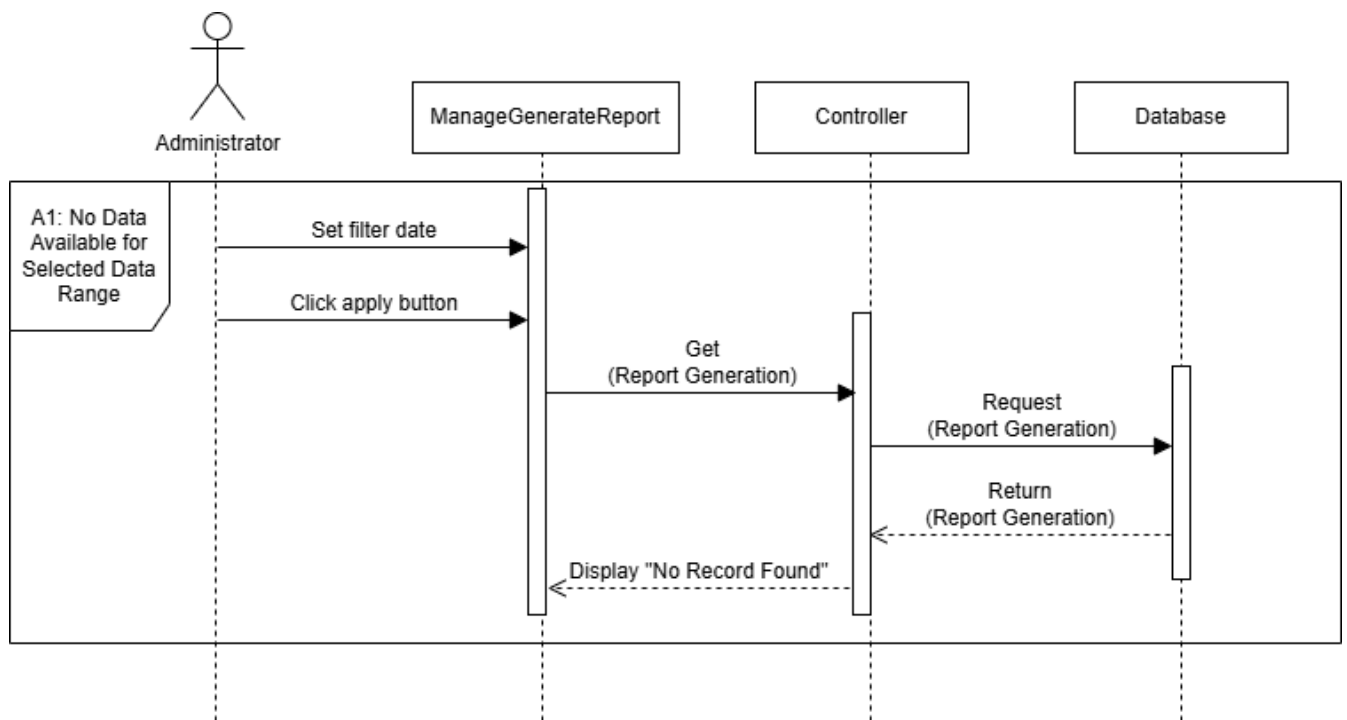


Figure 3.16 Alternative Flow Administrator for Manage Generate Report

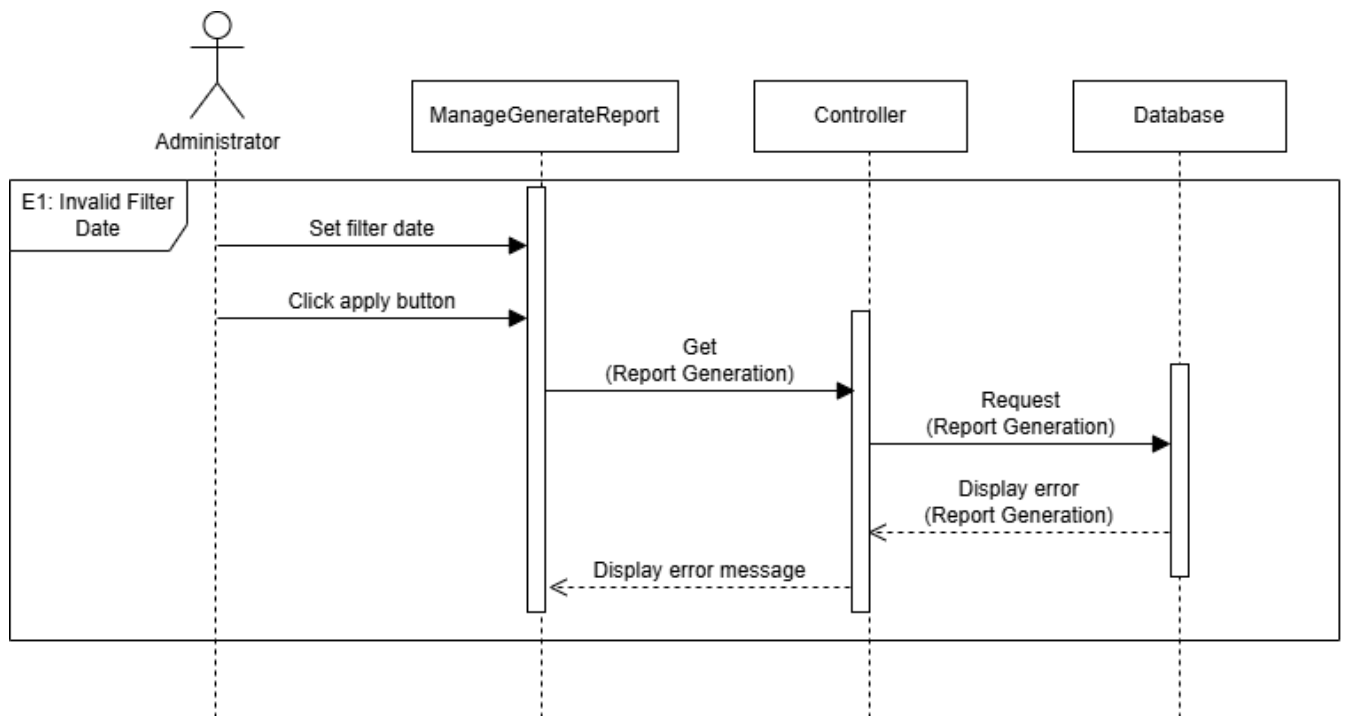


Figure 3.17 Exeption Flow Administrator for Manage Generate Report

3.2.3 Use Case Name User Interfaces

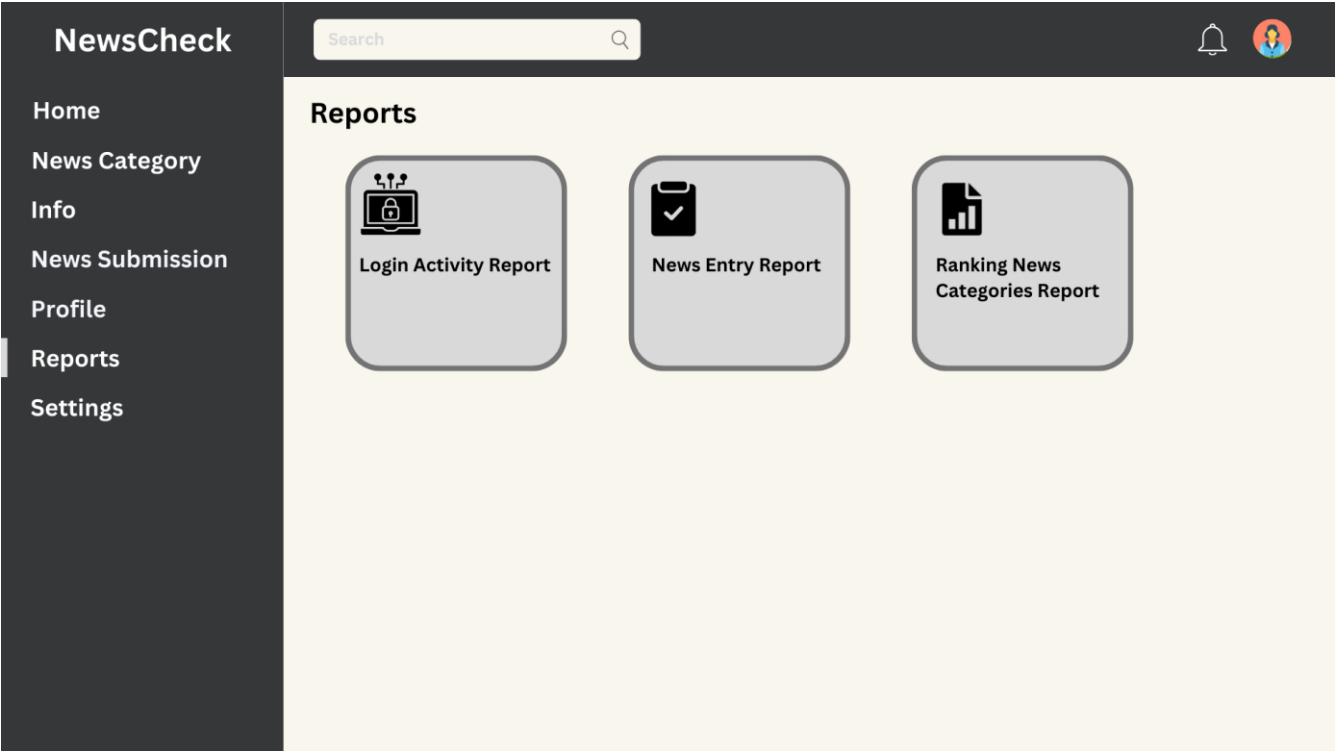


Figure 3.18 User Interface of Ranking News Categories Before Filter for Administrator

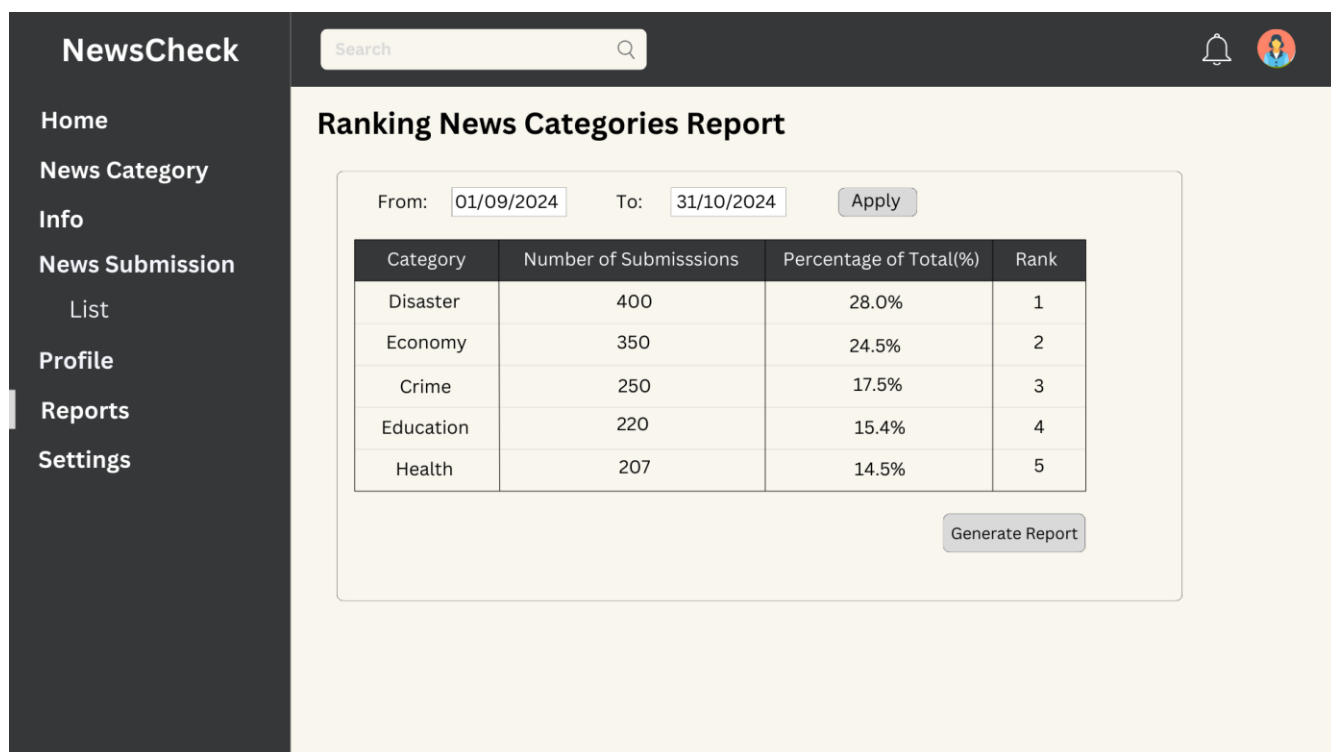


Figure 3.19 User Interface of Ranking News Categories Before Filter for Administrator

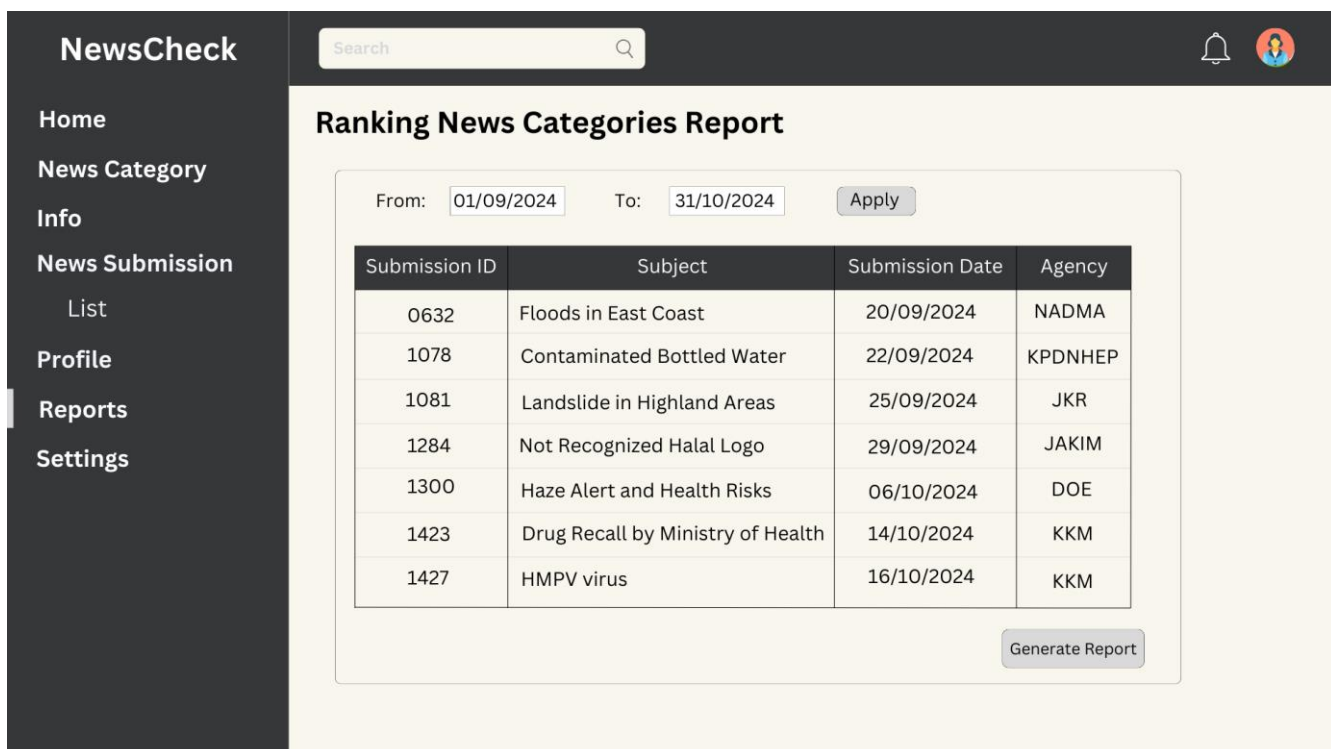


Figure 3.20 User Interface of Ranking News Categories List After Filter for Administrator

Table 3.4 Manage Generate Report - User Interface Description

User Interface Name	Description
News Submission List Before Filter for Administrator	The Agency can filter the date and generate report with click the generate report button of ranking news categories list in this interface.
News Submission List After Filter for Administrator	The Agency can view and generate report with click the generate report button of the ranking news categories list after filter the date and in this interface.

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 USER INTERFACES

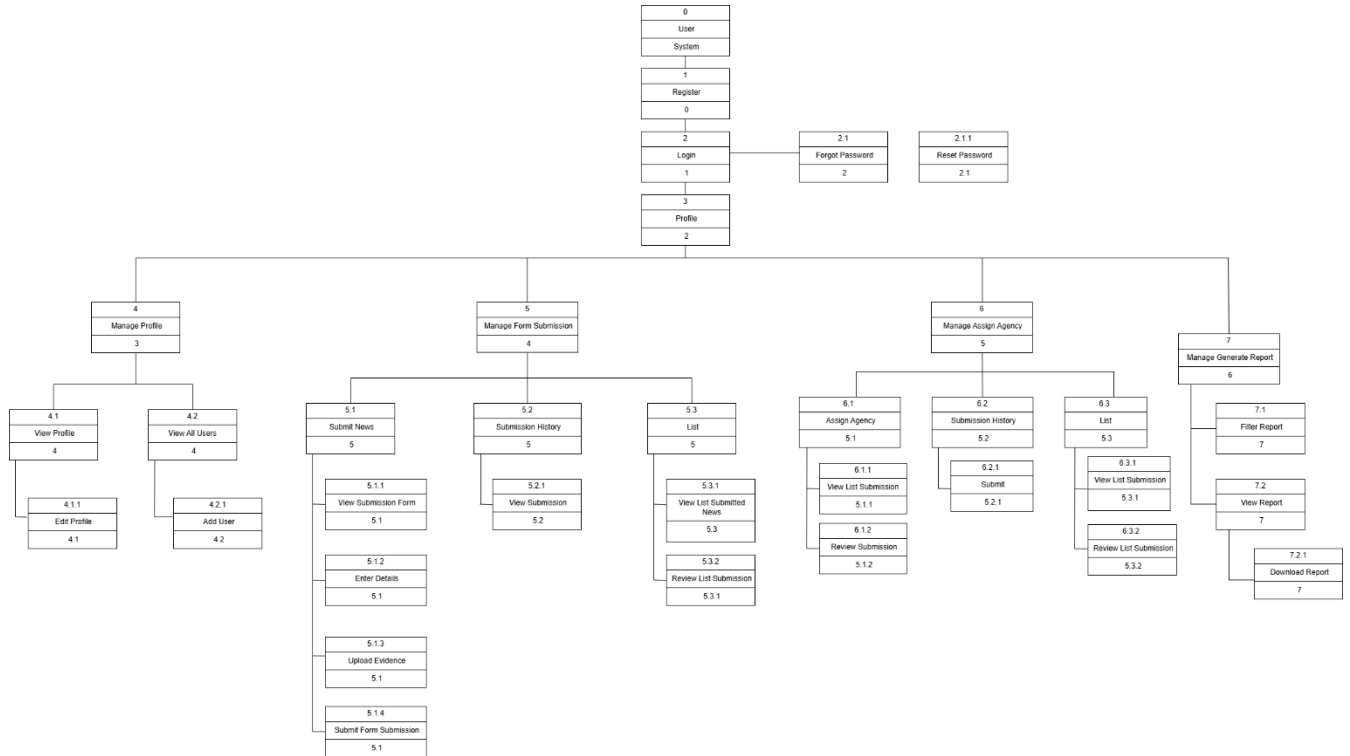


Figure 4.1 Dialog Diagram for News Check System

4.2 HARDWARE INTERFACES

Not applicable.

4.3 SOFTWARE INTERFACES

Table 4.1 Software Interfaces for News Check System

Application	API Interface and Descriptions
Chrome	- Browser API provided by https://developer.chrome.com/ - Chrome is a browser that can be used by the user to access the website.
MySQL Workbench 8.0 CE	Visual tool for database design provided by https://www.mysql.com/products/workbench

	• MySQL Workbench is used to store and manipulate data received from the system.
--	--

5. REQUIREMENT TRACEABILITY

Table 5.1 Requirement Traceability for Manage Assign Agency.

Requirement ID	Description	Interface ID
RQ401-NCS	The public user needs to navigates on the <<News Submission>> and <<Submission History>> to see the list of news submission.	Refer figure 3.8
RQ402-NCS	The administrator needs to navigates on the <<News Submission>> and <<List>> to see the list of news submission.	Refer figure 4.0
RQ403-NCS	The administrator needs to click on the <<View>> button to see the detail of news submission. It will show the detailed news that filled by public user.	Refer figure 4.0
RQ404-NCS	The administrator needs to click on the <<Submit>> button to forward the news to agency. It will to go back the list of news submission page.	Refer figure 4.1
RQ405-NCS	The management approver needs to navigates on the <<News Submission>> and <<List>> to see the list of news submission.	Refer figure 3.9
RQ406-NCS	The agencies need to navigates on the <<News Submission>> and <<List>> to see the list of news submission.	Refer figure 3.3
RQ407-NCS	The administrator needs to click on the <<Request Update>> button to return form to public user. It will to go back the list of news submission page.	Refer figure 4.1

Table 5.2 Requirement Traceability for Manage Generate Report.

Requirement ID	Description	Interface ID
RQ501-NCS	The administrator needs to navigates on the <<Reports>> section to see the type of report.	Refer figure 4.8
RQ502-NCS	The administrator needs filter the date and click on the <<Apply>> button to filter submissions. It will display the list of news submission after filter.	Refer figure 4.9
RQ503-NCS	The administrator needs to click on the <<Generate Report>> button to generate the report of list of news submission.	Refer figure 4.9-5.0

6. ACRONYMS AND ABBREVIATION

Terms	Definition
SRS	Software Requirement Specification
NCS	News Check System
RQ	Requirement
UC	Use Case